

REDES DALE CAMP, United Kingdom

WMO: 032300

Lat: 55.2850N

Lon: 2.2794W

Elev: 696

StdP: 14.33

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	24.4	18.6	14.4	23.5	21.3	16.4	26.5	33.3	45.5	30.4	44.0	3.0	0	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	73.1	62.9	69.7	60.9	66.7	59.2	64.2	71.0	62.4	68.0	60.7	65.1	8.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	84.0	66.9	60.0	79.3	64.9	58.5	75.0	63.2	29.7	71.2	28.4	68.1	27.1	65.2	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	22.6	19.5	DB	15.3	78.5	5.1	3.2	11.6	80.9	8.6	82.7	5.7	84.6	2.0	86.9
			WB	15.1	67.8	5.1	2.2	11.5	69.4	8.5	70.7	5.6	71.9	1.9	73.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	37.1	37.6	39.8	43.8	48.6	53.5	56.9	56.4	52.5	46.9	41.1	37.5	
	DBStd	8.81	5.28	5.26	5.22	5.02	5.25	4.34	4.29	4.15	4.26	5.07	5.61	6.51	
	HDD50	2166	400	348	317	194	89	14	1	2	20	122	271	388	
	HDD65	6931	865	767	781	636	509	345	254	268	374	562	718	852	
	CDD50	715	0	0	1	8	46	120	215	200	97	25	3	0	
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0	
	CDH74	69	0	0	0	0	2	8	36	22	1	0	0	0	
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0	
Wind	WSAvg	8.0	9.2	9.6	9.1	8.0	8.2	7.2	6.6	6.5	6.9	7.7	8.1	8.5	
Precipitation	PrecAvg	41.5	4.5	3.5	2.8	2.8	2.7	2.8	3.1	3.3	2.9	4.1	4.4	4.6	
	PrecMax	53.6	9.0	7.4	5.0	6.1	4.4	6.8	5.8	7.4	5.6	7.2	9.1	11.7	
	PrecMin	29.2	0.7	0.9	0.9	0.7	0.7	0.7	1.0	0.6	0.5	1.5	1.4	1.5	
	PrecStd	5.6	1.8	1.9	1.0	1.4	1.1	1.6	1.3	1.6	1.3	1.7	1.7	2.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	52.4	58.0	65.7	72.2	74.7	78.1	77.4	70.8	62.0	56.4	53.3	
		MCWB	49.9	48.6	48.9	54.5	60.6	63.2	65.1	64.0	62.2	57.2	54.3	51.4	
	2%	DB	49.0	49.5	52.9	59.9	67.2	70.1	73.8	70.8	66.1	58.8	53.6	50.8	
		MCWB	47.2	47.1	47.6	51.8	57.3	60.7	63.4	62.5	59.1	55.6	51.7	49.3	
	5%	DB	47.2	47.8	50.3	55.9	62.8	66.3	70.0	67.2	63.2	57.0	51.6	49.1	
		MCWB	45.5	45.5	46.5	49.3	54.9	58.0	61.6	60.4	57.6	54.5	49.7	47.5	
	10%	DB	45.4	46.1	48.3	53.2	58.9	63.2	66.3	64.5	60.7	55.0	49.8	47.1	
		MCWB	43.7	43.9	45.2	48.0	52.4	56.4	59.7	58.8	56.1	52.4	47.9	45.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	49.7	51.3	56.0	62.5	65.3	67.1	65.9	63.9	58.5	54.8	51.9
			MCDB	51.3	51.6	55.9	63.2	70.9	72.5	75.4	72.2	69.4	60.2	56.3	52.9
2%		WB	47.6	47.7	48.9	52.7	58.1	61.7	64.5	63.5	60.9	56.5	52.1	49.6	
		MCDB	48.8	49.3	51.6	58.3	65.2	68.1	71.5	69.2	64.3	58.2	53.2	50.7	
5%		WB	45.8	46.1	47.1	50.7	55.5	59.2	62.5	61.7	58.7	54.9	50.0	47.8	
		MCDB	47.1	47.6	49.7	55.1	61.3	64.7	68.6	66.1	61.7	56.8	51.3	49.1	
10%		WB	43.9	44.1	45.5	48.7	53.3	57.2	60.5	60.1	56.9	52.9	48.1	45.7	
		MCDB	45.3	45.9	48.1	52.6	58.3	61.9	65.4	63.6	59.9	54.7	49.5	47.1	
Mean Daily Temperature Range		5% DB	MDBR	8.8	10.6	12.6	15.8	16.8	16.1	16.2	15.6	15.2	12.8	10.6	9.7
			MCDBR	10.4	12.0	15.8	21.3	23.8	23.1	23.1	20.0	19.4	14.4	11.6	10.7
	MCWBR		9.9	10.1	11.2	14.1	15.2	14.7	13.9	12.4	13.1	11.5	10.6	10.5	
	5% WB	MCDBR	10.1	11.2	14.3	19.3	21.5	21.0	20.8	17.4	16.4	13.0	11.4	10.6	
		MCWBR	10.0	10.1	11.1	13.4	14.2	14.2	13.2	11.6	12.2	11.1	10.8	10.5	
Clear-Sky Solar Irradiance	taub	0.310	0.325	0.355	0.377	0.376	0.376	0.376	0.366	0.349	0.333	0.308	0.293		
	taud	2.284	2.372	2.311	2.279	2.323	2.347	2.350	2.390	2.447	2.447	2.354	2.254		
	Ebn at Noon	193	239	257	267	275	276	273	269	258	233	194	169		
	Edh at Noon	18	23	31	36	37	37	36	32	27	21	16	15		
All-Sky Solar Radiation	RadAvg	174	377	719	1166	1521	1492	1436	1170	836	457	223	129		
	RadStd	24	39	77	121	107	140	147	89	69	51	20	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page